

Climate 5000 Series - Gen 4

12K BTU Single Zone Max Performance System

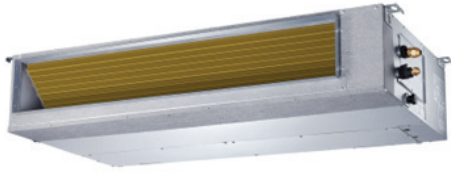
Ducted IDU

Engineering
 Submittal
 Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

12K Btu Single Zone Max Performance System - Ducted IDU

Indoor part number		8-733-962-710
Indoor model number		BMS500-AAU012-1AHDXD
Outdoor part number		8-733-962-689
Outdoor model number		BMS500-AAS012-1CSXHD
Efficiency Data		AHRI 210/240 2023
SEER2		20.0
EER2		12.0
HSPF2 Region IV		10
HSPF2 Region V		8
COP(47F)		3.39
COP (5F)		1.90
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3400~13500
	Rated Capacity	12000
	Current	4.6
	Input Power	1000
Heating	Capacity Range	3800~16200
	Rated Capacity (47F)	12500
	Rated Capacity (5F)	11800
	Current	5.0
	Input Power	1080
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 15.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 350
Input (W)		885
Rated current (RLA/A)		6.0
Piping		
Refrigerant precharge (oz/kg)		35.27 / 1.00
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82.00 / 25
Max. difference in height (ft/m)		32.80 / 10

1 of 7

Operation Range

ODU	Cooling (°F / °C)	-22 ~ 122 / -30 ~ 50
	Heating (°F / °C)	-22 ~ 75 / -30 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		N/A
IDU	Fan motor model	ZKFN-81-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	150.0
	Fan speed, H / M / L (RPM)	1070 / 990 / 910
	Air flow, Turbo / H / M / L / Si (CFM)	400.2 / 323.7 / 264.9 / 176.6 / 117.7
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 36 / 34.5 / 32.5 / 25.5
	Net weight (lbs / kg)	39.46 / 17.9
	Gross weight (lbs / kg)	46.74 / 21.2
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed, H / / L (RPM)	780 / 680 / 580
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	56.0
ODU	Net weight (lbs / kg)	72.75 / 33
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK2-7X1 (8733951011)	25ft Lineset

Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

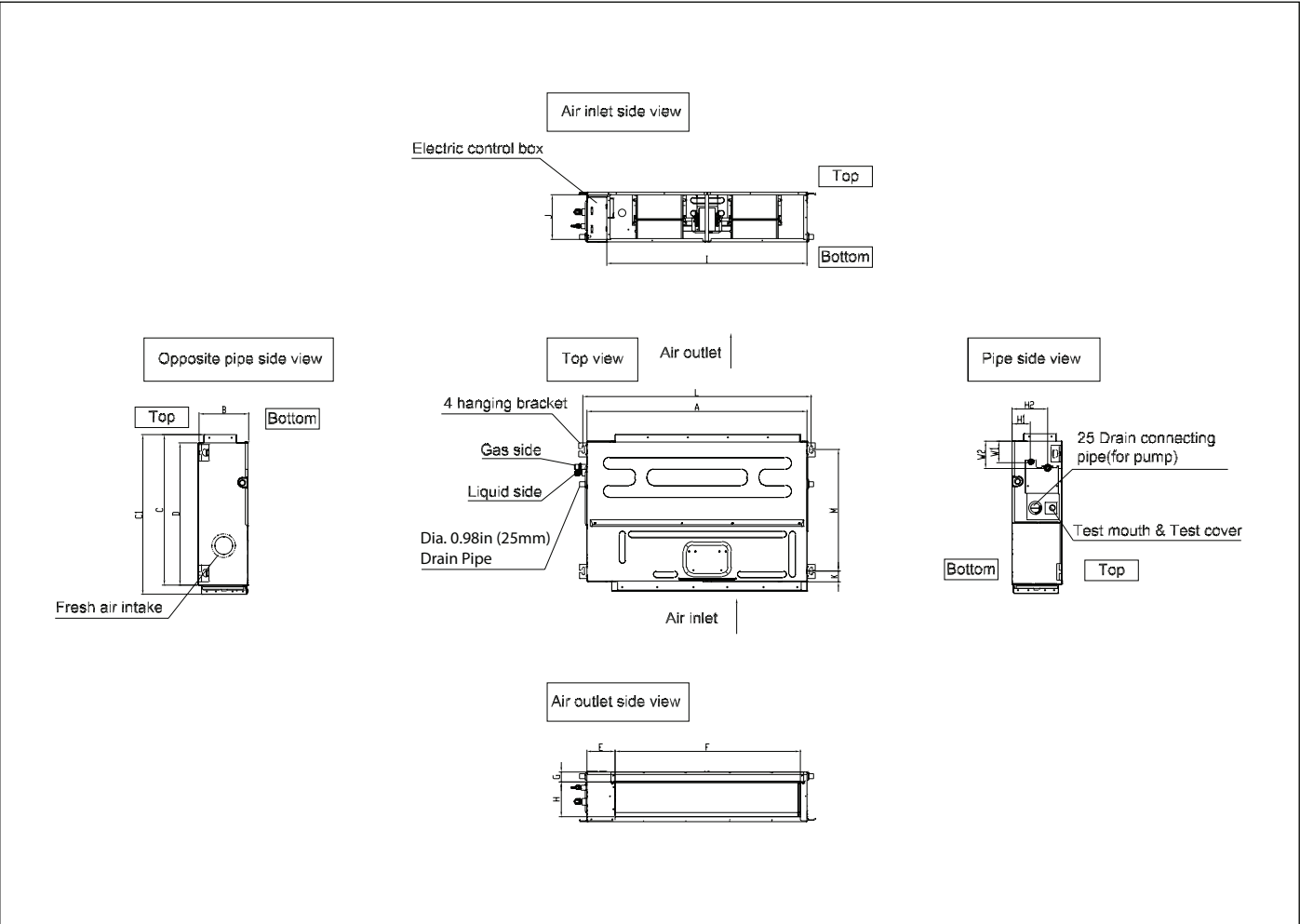
Tel: 1-800-283-3787 www.bosch-homecomfort.us

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12K Btu Single Zone Max Performance System - Ducted IDU



Indoor Unit Dimensions and Connections

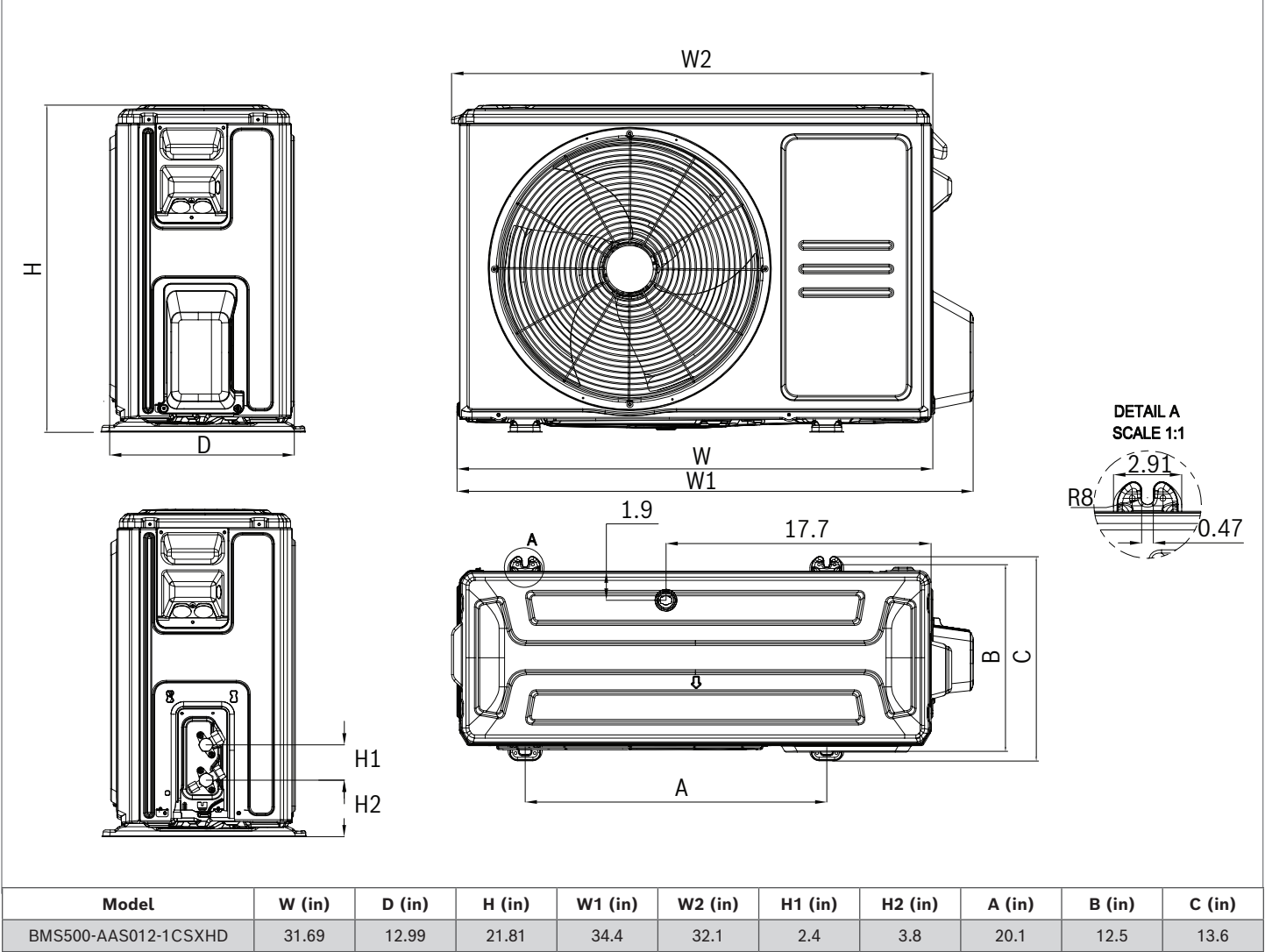


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU012-1AHDXD	27.6	7.9	19.9	17.7	5.4	21.1	1.2	6.0	23.6	7.3	2.0	29.2	14.2	3.3	5.5	3.3	3.3

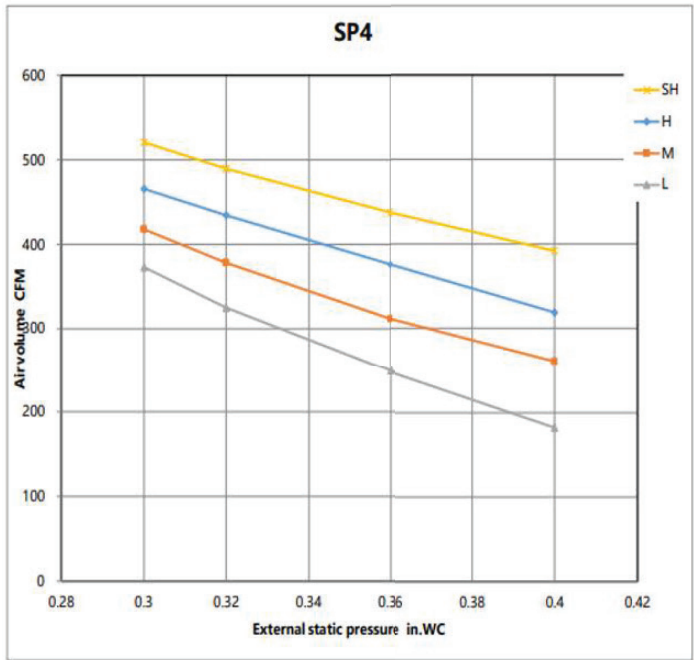
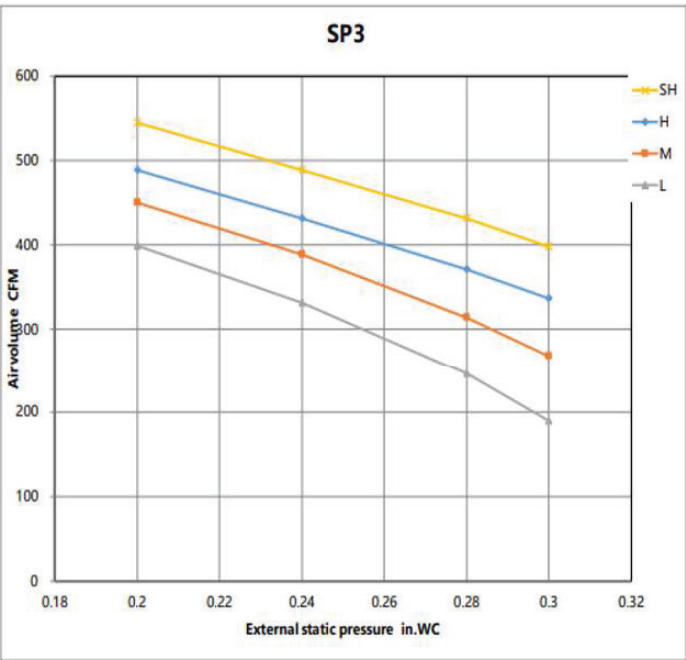
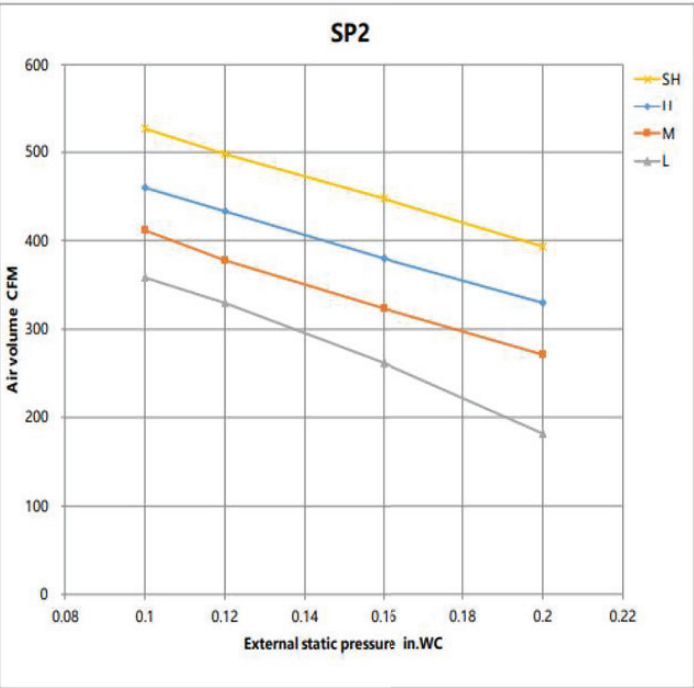
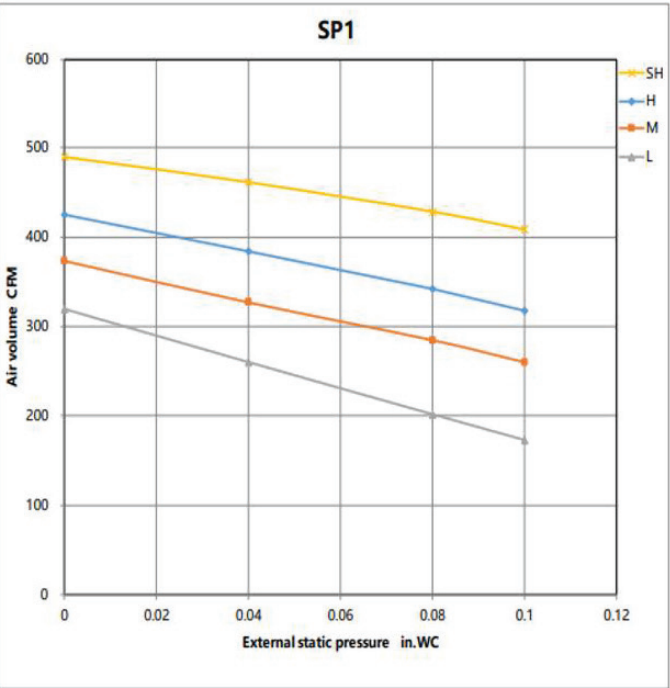
Climate 5000 Series - Gen 4
12K Btu Single Zone Max Performance System - Ducted IDU



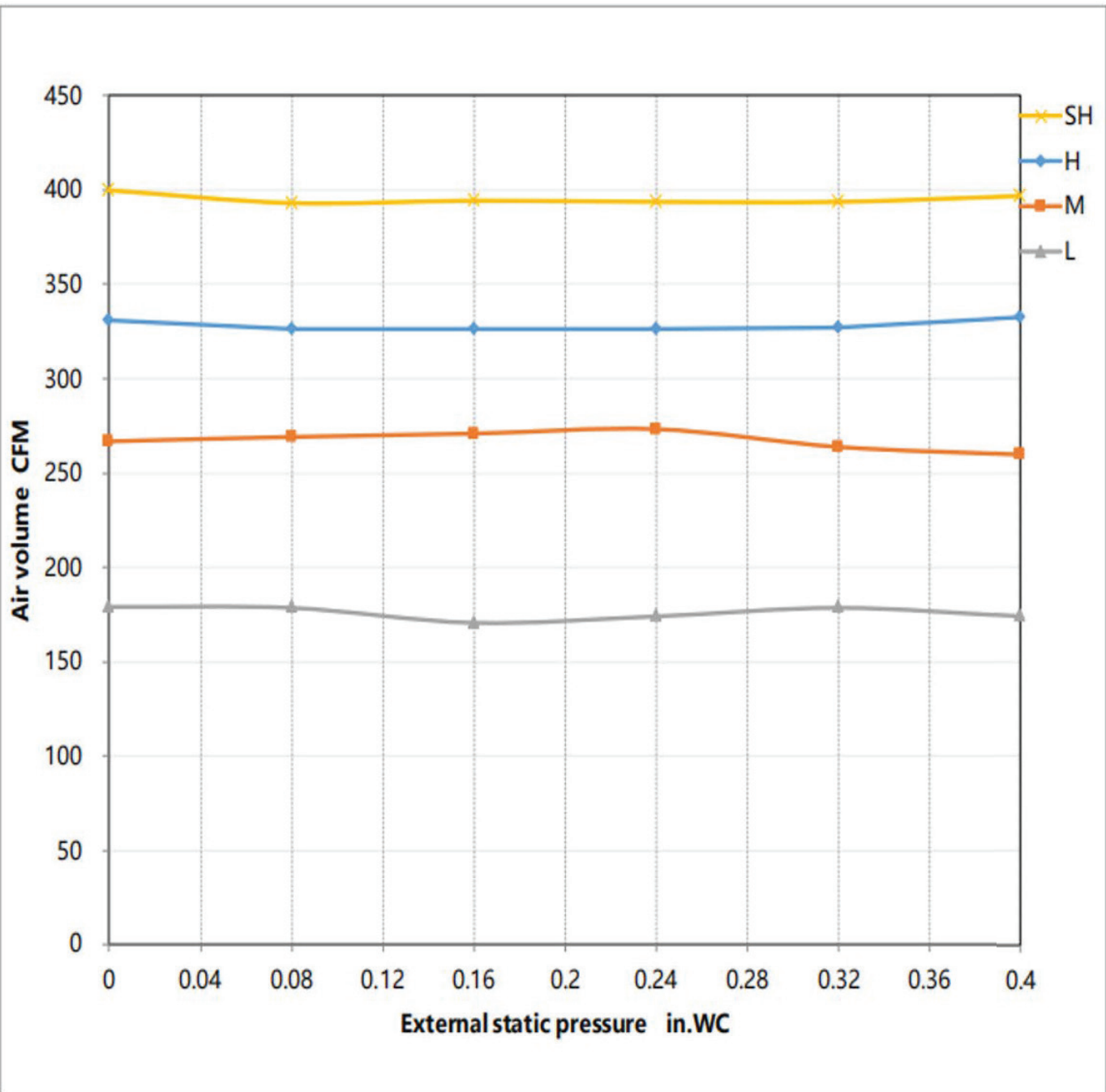
Outdoor Unit Dimensions and Connections



Air Volume



Air Volume



Climate 5000 Series - Gen 4

12K Btu Single Zone Max Performance System - Ducted IDU


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Cooling Performance - Indoor Dry Bulb Temperature at 400 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	9916	9911	10053	10095	10372	10346	10337	10331	10330	10321	10655	10642	10632	10630	10620	11239	11236	11213	11209	11185
	S/T	0.80	0.91	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67
	PI	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
-13°F (-25.0°C)	TC	10742	10737	10891	10936	11237	11208	11199	11192	11191	11181	11543	11529	11518	11516	11505	12176	12173	12148	12143	12117
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
-4°F (-20.0°C)	TC	11569	11563	11728	11777	12101	12070	12060	12053	12052	12042	12431	12416	12404	12402	12390	13112	13109	13082	13077	13049
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.62	0.62	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62
0°F (-17.8°C)	TC	11953	11946	12167	12218	12554	12465	12451	12442	12440	12566	12832	12813	12798	12796	12790	13522	13519	13517	13519	13528
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.63	0.63	0.63	0.63	0.63
5°F (-15.0°C)	TC	12295	12338	12620	12678	13069	12872	12875	12876	12889	13081	13284	13287	13290	13290	13308	14063	14058	14066	14067	14074
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.65	0.65	0.65	0.65	0.65
14°F (-10.0°C)	TC	11573	11567	11803	11858	12226	12147	12136	12128	12127	12240	12563	12548	12535	12533	12520	13359	13355	13329	13324	13296
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71
17°F (-8.3°C)	TC	11306	11343	11605	11661	12040	11869	11874	11876	11886	12053	12274	12278	12283	12284	12299	13053	13050	13048	13050	13061
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.74	0.74	0.73	0.73	0.73	0.73	0.73	0.73	0.73	0.73	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74	0.74
23°F (-5.0°C)	TC	11766	11817	12123	12182	12584	12362	12366	12367	12383	12597	12788	12793	12796	12796	12817	13598	13594	13603	13605	13613
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.77	0.77	0.77	0.77	0.77
35°F (1.7°C)	TC	12804	12805	12880	12916	13332	13403	13402	13403	13403	13439	13822	13821	13822	13822	13824	14601	14597	14596	14596	14597
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.71	0.71	0.72	0.72	0.75	0.76	0.76	0.76	0.76	0.76	0.79	0.79	0.79	0.79	0.79	0.86	0.86	0.86	0.86	0.86
41°F (5.0°C)	TC	12367	12367	12466	12552	12924	12942	12942	12943	12943	12988	13346	13345	13346	13346	13347	14095	14092	14091	14090	14092
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.65	0.65	0.66	0.67	0.69	0.70	0.70	0.70	0.70	0.70	0.73	0.73	0.73	0.73	0.73	0.79	0.79	0.79	0.79	0.79
47°F (8.3°C)	TC	11930	11937	12098	12152	12511	12483	12482	12483	12482	12540	12870	12869	12870	12870	12870	13591	13587	13586	13586	13587
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.60	0.60	0.61	0.61	0.64	0.64	0.64	0.64	0.64	0.64	0.67	0.67	0.67	0.67	0.67	0.73	0.73	0.73	0.73	0.73
50°F (10.0°C)	TC	11714	12036	12443	12515	13000	12375	12382	12468	12507	13012	12858	12875	12881	12882	13006	13744	13780	13816	13820	13837
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67
59°F (15.0°C)	TC	11708	12118	12548	12625	13138	12387	12404	12545	12638	13151	12898	12911	12916	12934	13162	13850	13878	13905	13909	13920
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.69	0.69	0.69	0.69	0.69
68°F (20.0°C)	TC	11696	12195	12647	12727	13267	12382	12410	12661	12741	13281	12917	12927	12950	12961	13292	13918	13944	13966	13968	13975
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.69	0.69	0.69	0.69	0.70	0.69	0.69	0.69	0.69	0.69	0.70	0.70	0.70	0.70	0.70	0.71	0.71	0.71	0.71	0.70
77°F (25.0°C)	TC	11674	12236	12715	12800	13372	12376	12415	12729	12814	13385	12936	12951	12984	12999	13396	13972	14014	14047	14050	14063
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.70	0.70	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.71	0.72	0.72	0.72	0.72	0.72
82°F (27.8°C)	TC	11730	12328	12826	12913	13504	12430	12478	12837	12927	13518	13008	13024	13065	13081	13530	14064	14117	14154	14158	14173
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.77	0.77	0.77	0.77	0.78	0.77	0.77	0.77	0.77	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78
86°F (30.0°C)	TC	11900	12494	13005	13096	13706	12550	12607	13019	13111	13720	13144	13159	13208	13227	13732	14230	14285	14319	14322	14336
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.89	0.90	0.90	0.90	0.91	0.90	0.90	0.90	0.90	0.91	0.90	0.90	0.90	0.90	0.91	0.92	0.91	0.91	0.91	0.91
95°F (35.0°C)	TC	12195	12261	12658	12733	13236	12961														

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Heating Performance - Indoor Dry Bulb Temperature at 400 CFM										
Outdoor DB(°F)	TC□Total Heating Capacity(BTU/hr)					PI□Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	6606	6474	6442	6416	6358	1.46	1.47	1.47	1.47	1.48
-13°F(-25.0°C)	8324	8158	8117	8085	8012	1.53	1.54	1.55	1.55	1.55
-4°F(-20.0°C)	10518	10308	10256	10215	10124	1.64	1.65	1.66	1.66	1.66
0°F(-17.8°C)	11221	10997	10942	10898	10800	1.71	1.72	1.73	1.73	1.74
5°F(-15.0°C)	12100	11858	11799	11752	11646	1.80	1.82	1.82	1.82	1.83
14°F(-10.0°C)	13142	12879	12815	12763	12648	1.88	1.90	1.90	1.90	1.91
17°F(-8.3°C)	13489	13219	13153	13101	12983	1.91	1.92	1.93	1.93	1.94
22°F(-5.6°C)	13790	13515	13447	13393	13273	1.85	1.86	1.87	1.87	1.88
27°F(-2.8°C)	14092	13810	13741	13686	13563	1.79	1.80	1.80	1.81	1.82
32°F(0.0°C)	14393	14105	14035	13979	13853	1.73	1.74	1.74	1.75	1.75
35°F(1.7°C)	14574	14282	14211	14154	14027	1.69	1.70	1.71	1.71	1.72
37°F(2.8°C)	14909	14610	14537	14479	14349	1.64	1.66	1.66	1.66	1.67
42°F(5.6°C)	15745	15431	15353	15292	15154	1.53	1.54	1.54	1.55	1.55
44.6°F(7.0°C)	16080	15759	15680	15617	15477	1.48	1.50	1.50	1.50	1.51
47°F(8.3°C)	16582	16251	16169	16105	15960	1.41	1.43	1.43	1.43	1.44
52°F(11.1°C)	16381	16054	15973	15909	15766	1.27	1.28	1.28	1.29	1.29
57°F(13.9°C)	16180	15857	15777	15714	15573	1.13	1.14	1.14	1.14	1.15
59°F(15.0°C)	16100	15778	15699	15636	15495	1.07	1.08	1.08	1.09	1.09
62°F(16.7°C)	15979	15659	15581	15519	15379	0.99	1.00	1.00	1.00	1.00
64.4°F(18.0°C)	15883	15565	15487	15425	15286	0.92	0.93	0.93	0.93	0.93
69.8°F(21.0°C)	13573	13302	13235	13182	13064	0.77	0.77	0.77	0.78	0.78
75.2°F(24.0°C)	10105	9903	9854	9814	9726	0.61	0.62	0.62	0.62	0.62

Bolded values were tested. All other values are simulated and for reference only.